Matric2Uni

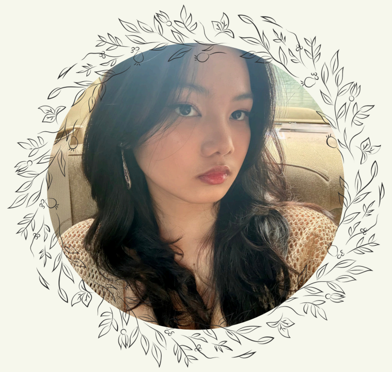
From Marks to Your Dream University

Empowering Students for a Future-Ready Career

GROUP Members

Ye Yint Hein
TNT- 2329

Phyo Pyae Pyae Khaing
TNT-2464



Htet Nay Chi Ko
TNT-2408

Hsu Wai Phyo
TNT-2373

Shoon Lae Aung
TNT-2349

01

Outlines

01 Introduction

02 Objective

03 Project Functions

04 C++ Concepts Used

05 Conclusion

Introduction

- **Program Name: Myanmar Universities Recommendation System**
- Project Idea:

A C++ program that recommends universities in Myanmar to students based on their STEAMS exam marks.

- Motivation:

Many students are unsure which universities they are eligible for after the matriculation exam.

- Usefulness:

The system helps students quickly identify suitable universities, saving time and guiding career decisions.

Objectives

- To provide a recommendation system that checks eligibility for Myanmar universities.
- To support students by matching their marks with university requirements.
- To improve decision-making by giving clear and fast results about possible admissions.

Program Features

- Welcomes user by name.
- Allows STEAMS (3 options) selection.
- Accepts marks input (valid range 40–100).
- Shows marks summary and eligibility rules on request.
- Lets user select a specific region (or search all).
- Filters by field of interest (Medical, IT, Engineering, Economics, Education, Language, etc.).
- Displays eligible universities with clear requirements.
- Repeats process if user wants to try again.

Functions Used in main()

1. clearScreen()

• Clears console screen (Windows & Linux support)

2. getValidatedName(prompt)

• Gets name input

• Validates: not empty, only letters & spaces, ≤ 30 chars

3. class User

• Constructor → greet user

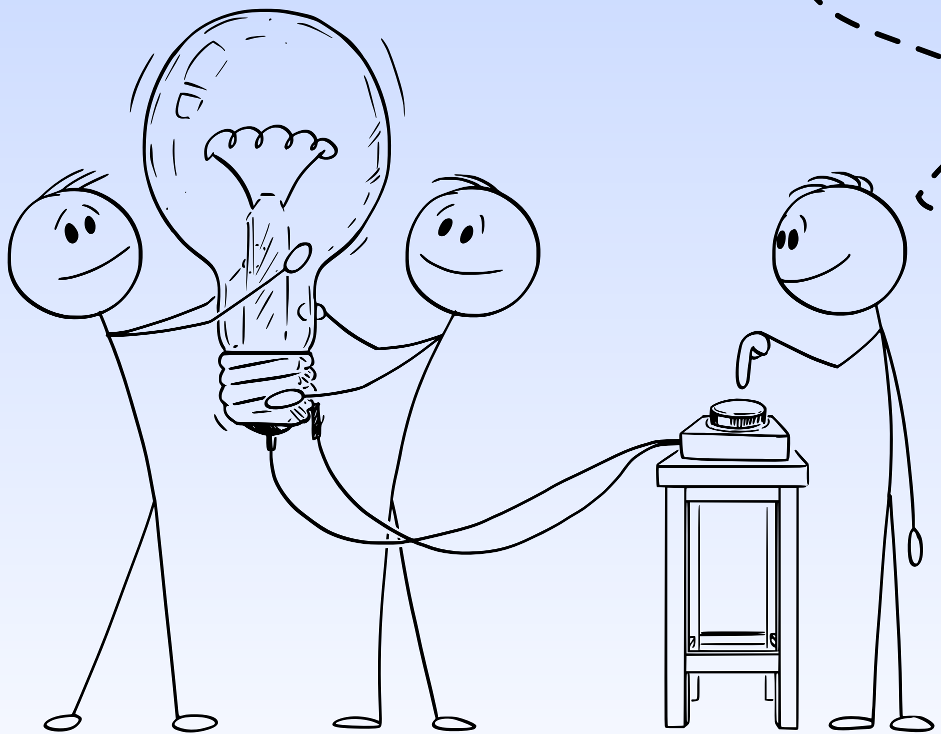
• Destructor → thank user at program exit

• getName() → return username

4. aboutProgram()

• Shows purpose of program

• Explains flow (steps 1–4)



5. showMarksSummary(...parameters)

• Displays calculated marks summary

• Shows total marks, English+Math, Science combinations

🎓 UNIVERSITY ELIGIBILITY SYSTEM – CORE LOGIC

We use a struct called University to represent each university in Myanmar.

```
struct University {
    string name;
    string shortName;
    bool isMedical;
    bool isSpecial; //   YTU, MTU,...
    bool isComputer;
    bool isPU;  // Polytechnic Uni
    string field;
    int minTotal;
    int engMathMin;
    int engMathPhysChemMin;  };
```

bool isEligible(const University &u, int totalMarks, int engMathTotal, bool hasBiology, int engMathPhysChemTotal, int steamsChoice) {

```
⚙️ Special Universities (YTU, MTU, ...)
if (u.isSpecial)
  return (totalMarks >= u.minTotal &&
      engMathPhysChemTotal >= u.engMathPhysChemMin);
💻 IT / Computer Universities (UCS, CU)
if (u.isComputer)
  return (totalMarks >= u.minTotal || engMathTotal >=
u.engMathMin);
```

```
🏥 Medical Universities
if (u.isMedical)
   return (totalMarks >= u.minTotal && hasBiology);
🎓 Yangon University of Education (YUOE)
if (u.shortName == "YUOE") {
   if (steamsChoice == 1) return totalMarks >= 380;
   if (steamsChoice == 2) return totalMarks >= 360;
   return totalMarks >= 350; // STEAMS 3
}
```

```
🏗️ Polytechnic Universities (PU)
if (u.isPU && u.shortName == "NSPU")
  return (totalMarks >= u.minTotal ||
       engMathTotal >= u.engMathMin ||
       engMathPhysChemTotal >= u.engMathPhysChemMin);
if (u.isPU)
  return (totalMarks >= 320 ||
       engMathPhysChemTotal >= u.engMathPhysChemMin);
📚 General Universities
   return totalMarks >= u.minTotal;
```

🎓 UNIVERSITY ELIGIBILITY SYSTEM – CORE LOGIC

Regions & Universities Dataset

----In Myanmar, universities are grouped by 15 regions and states.----

🗺️ Regions included:

```
string locations[15] = {
        "Yangon Region", "Mandalay Region", "Ayeyarwady Region", "Bago Region",
        "Magway Region", "Sagaing Region", "Tanintharyi Region",
        "Kachin State", "Kayah State", "Kayin State", "Chin State",
        "Mon State", "Rakhine State", "Shan State", "Nay Pyi Taw",
    };
```

🏫 Each region has a list of universities.

For example:

- Yangon Region: University of Yangon, Dagon University, YTU, UIT, UCSY, University of Medicine (1 & 2), etc.
- Mandalay Region: University of Mandalay, MTU, UTYCC, UCSM, University of Medicine (Mandalay), etc.
- Other Regions: Each has a mix of General, Computer, Technological, Medical, and Educational colleges.

👉 This dataset is stored as a 2D array universities[15][30]

- 15 rows = Regions.
- Up to 30 entries per region = Universities.
- This lets the program show:
- All universities in a chosen region, or
- Only eligible universities in that region.

// Regions (15 states/regions + Nay Pyi Taw)

```
string locations[15] = {
  "Yangon", "Mandalay", "Ayeyarwady", "Bago", "Magway", "Sagaing",
  "Tanintharyi", "Kachin", "Kayah", "Kayin", "Chin",
  "Mon", "Rakhine", "Shan", "Nay Pyi Taw"
};

// Universities dataset: [region][universities]
University universities[15][30] = {
  { // Yangon Region
    {"University of Yangon","YU",false,false,false,false,"General",240,0,0},
    {"Yangon Technological University","YTU",false,true,false,false,"Engineering",240,0,300},
    {"University of Medicine 1","UM1",true,false,false,false,"Medical",450,0,0},
    {"University of Computer Studies (Yangon)","UCSY",false,false,true,false,"IT",450,140,0},
    {"University of Information Technology","UIT",false,true,false,false,"IT",450,0,0},
    {"Yangon University of Education","YUOE",false,false,false,false,"Education",0,0,0}
  },

  { // Mandalay Region
    {"University of Mandalay","UMdl",false,false,false,false,"General",240,0,0},
    {"Mandalay Technological University","MTU",false,true,false,false,"Engineering",240,0,300},
    {"UTYCC","UTYCC",false,true,false,false,"Engineering",420,0,280}
  },

  { // Ayeyarwady Region
    {"Pathein University","PU",false,false,false,false,"General",240,0,0},
    {"UCS (Pathein)","UCS-Pth",false,false,false,false,"IT",320,0,0},
    {"TU (Pathein)","TU-Pth",false,true,false,false,"Engineering",240,0,240},
    {"Polytechnic Univ. (Maubin)","Poly-Mb",false,false,false,true,"Engineering&IT",320,0,240}
  },

  // ... other regions (Bago, Magway, Sagaing, Shan, Nay Pyi Taw, etc.)  };
```

Function in main()

getInt(prompt, min, max)

- Gets an integer input from the user.
- Ensures the number is within the valid range (e.g., 1–4 for menu).

clearScreen()

- Clears the console screen.
- Keeps output clean and easy to read.

aboutProgram()

- Displays information about the program's purpose.

runOriginalFlow(name)

- Runs the main workflow of the program.
- Checks eligibility and recommends universities.

getValidatedName(prompt)

- Asks the user for their name.
- Validates the input (no numbers, not empty, etc.).

User currentUser(name)

- Creates a user object with the give name.
- May show a welcome message.

getYesNo(prompt)

- Asks the user a Yes/No question.
- Returns 'y' or 'n' for decision-making.

What Function Are Used?

Types	Function	Purposes
Constructor	User::User()	Ask name & show welcome banner
Destructor	User::~User()	Return username
Getter	User::getName()	Show farewell banner
Free Function	showMarksSummary(...	Displays total marks and subject breakdown.
	showEligibilityRules()	Explain admission rules
	isEligible(...)	Check if student qualifies
	aboutProgram()	Explains the program purpose and flow.
Entry Point	main()	Program driver; menu for navigation.
	runOriginalFlow	Main Workflow (input,eligibility,check,region/fiel

What C++ Concepts are used in this Program?

- Looping (for / while): Iterate through universities to compare marks.
- Conditional Statements (if-else): Decide eligibility based on exam marks.
- Input/Output (cin, cout): Interaction with the user.
- Structure: Used to represent university details (name, short name, requirements, fields).
- Arrays: Store multiple universities and allow iteration for eligibility checks.
- Functions: Separate modules for program description, input, and recommendation.

Conclusion

Project Achievements:

- Successfully developed a system that recommends universities based on student marks.
- Program works for medical, IT, engineering, and other fields.

Benefits:

- Saves students time and effort.
- Provides fast and reliable eligibility checking.
- Simple and user-friendly interface.

Future Improvements:

- Add file handling to save student results.
- Develop a graphical interface for better usability.
- Include detailed university information (location, departments, etc.).

Thank You!!

Wishing you success ahead.